SmartRouteADMM - AI driven Scheduling and Route Management System using ADMM

TEAM - 07

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Table Of Contents

1. Introduction 3
2. Objective 4
3. Problem Statement 5
4. Why This Problem Is Important? 6-7
5. Project Scope 8-9
6. WorkFlow 10
7. Methodology 11-14
8. Initial Research and Work Done 15
9. Challenges and Risk 16
10. Next Steps 17
11. Conclusion 18
12. Literature Review/ References 19

Introduction

Public transportation systems face challenges such as static schedules, inefficient routes, and high operational costs.

- Increasing urban congestion and environmental concerns demand smarter transit solutions.
- Advances in AI, IoT, and GIS now enable dynamic, real-time scheduling and route optimization.
- Traditional bus scheduling methods do not respond to real-time variables (e.g., traffic, passenger demand).
- This project explores an integrated system that leverages real-time data to adjust bus routes and schedules dynamically.

Objective

Optimize Efficiency:

- Implement data-driven route planning to minimize passenger wait times.
- Reduce unnecessary bus trips and idle time to improve fleet utilization.

Enhance Adaptability:

- Integrate real-time traffic and demand data to dynamically adjust bus schedules.
- Utilize AI/ML algorithms to predict peak and off-peak hours for optimized operations.

Promote Sustainability:

- Reduce fuel consumption by optimizing routes and minimizing empty bus runs.

Problem Statement

Public transportation systems across various cities and regions face challenges such as long passenger wait times, inefficient fleet utilization, and escalating operational costs—all of which contribute to environmental concerns through increased fuel consumption and emissions. Traditional static bus schedules and fixed routes are ill-equipped to respond to dynamic variables like fluctuating passenger demand, varying traffic conditions, and unexpected disruptions. This leads to suboptimal service reliability and decreased passenger satisfaction.

Why This Problem is Important ?

Enhanced Passenger Experience:

- **Reduced Wait Times:** Dynamic scheduling ensures buses arrive based on real-time demand, minimizing long waits.
- **Increased Service Reliability:** Adapting routes based on live traffic conditions prevents unexpected delays.

Cost Efficiency:

- **Lower Fuel and Maintenance Costs:** Optimized routes reduce unnecessary mileage, cutting fuel expenses and vehicle wear and tear.

- **Improved Resource Allocation:** Fleet utilization is maximized by aligning bus availability with demand, preventing empty or overcrowded trips.

Environmental Impact:

- **Lower Carbon Emissions:** Reduced fuel consumption directly decreases greenhouse gas emissions, promoting cleaner air.
- **Sustainable Urban Mobility:** Efficient public transit encourages more people to use buses instead of personal vehicles, reducing congestion and pollution.

Project Scope

Inclusions (What's Covered)

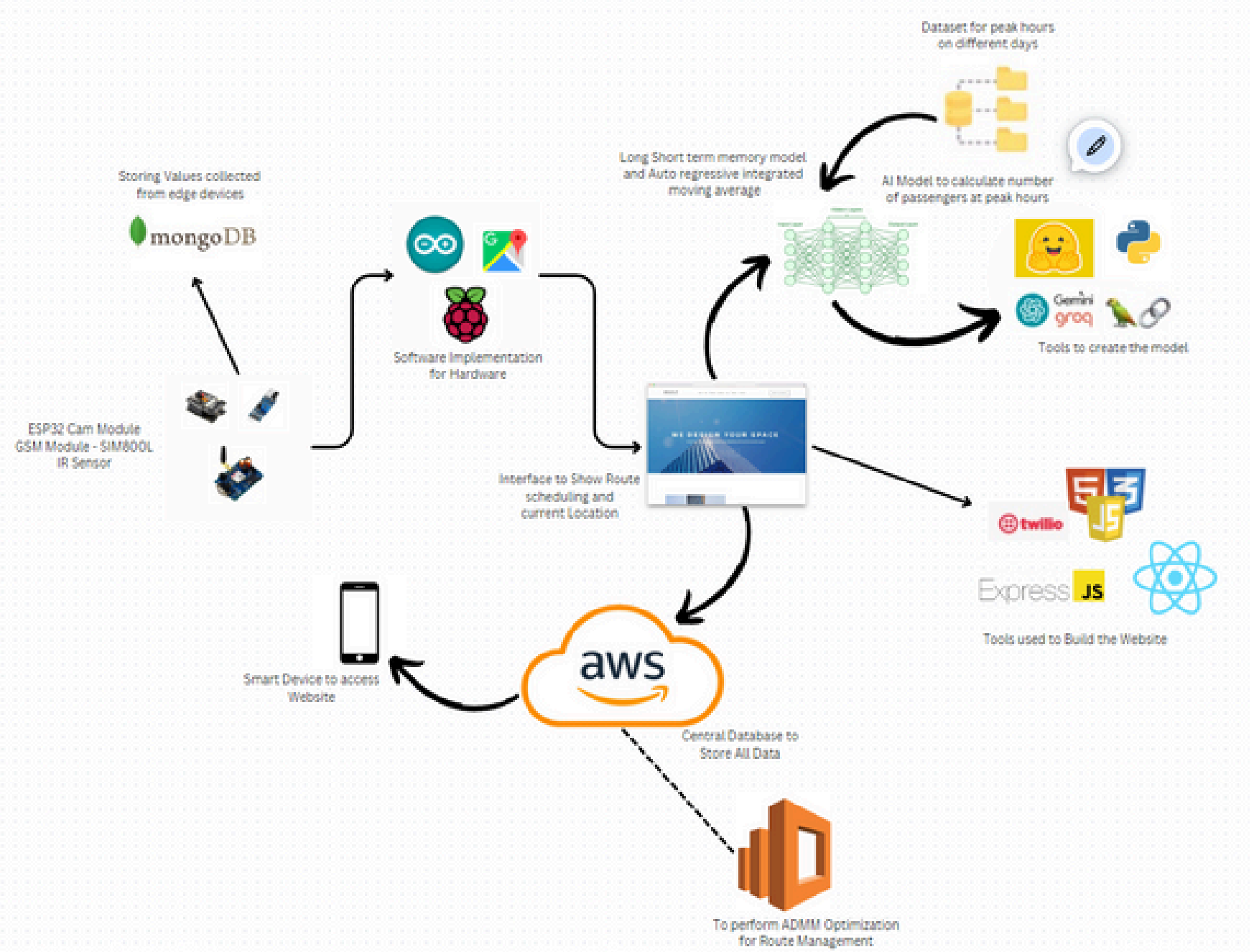
- **Real-Time Dynamic Scheduling:** Implement AI/ML-based route and schedule optimization.
- **Live Traffic & Demand Integration:** Adjust bus schedules based on real-time traffic and passenger demand.
- **Fleet Utilization Optimization:** Improve vehicle allocation to reduce idle buses and overcrowding.
- **Cost & Environmental Impact Reduction:** Optimize fuel consumption and lower operational costs.
- **Passenger Experience Enhancement:** Ensure better service reliability, reduced wait times, and improved accessibility.

Project Scope

Exclusions (What's Not Covered)

- **Infrastructure Development:** No changes to existing bus stops or new transportation infrastructure.
- **Fleet Expansion:** The project focuses on optimizing the current fleet, not purchasing new vehicles.
- **Fare Pricing & Ticketing Systems:** Does not include modifications to fare structures or payment methods.

WorkFlow


Dataset for peak hours on different days
Long Short term memory model and Auto regressive integrated moving average
AI Model to calculate number of passengers at peak hours
Storing Values collected from edge devices
mongoDB
Software Implementation for Hardware
Tools to create the model
ESP32 Cam Module
GSM Module - SIM800L
IR Sensor
Interface to Show Route scheduling and current Location
twilio
Express JS
Tools used to Build the Website
Smart Device to access Website
aws
Central Database to Store All Data
To perform ADMM Optimization for Route Management

Methodology

- **Data Collection:** Gather historical and real-time data from IoT sensors installed on buses.
- **Algorithm Development:** Develop AI models (using ADMM optimization) to dynamically adjust schedules and routes.
- **GIS Integration:** Use geographic information systems to analyze and optimize routes.
- **Edge & Cloud Processing:** Employ edge devices for local data preprocessing; utilize cloud platforms for central analytics and dashboarding.

Hardware Components

- IR Break Beam Sensor(TCRT5000)
- ESP32 Cam Module (Wifi/Bluetooth)

Factor	Details
Protocol	802.11 b/g/n (2.4 GHz)
Max Speed	150 Mbps (802.11n)
Real-World Speed	3–30 Mbps, depending on network congestion and range
Latency	5–30 ms (low latency, best for real-time data)
Range	Up to 100m (Open Space), ~50m Indoors
Best Use Case	Streaming camera feed, real-time passenger count to cloud

Factor	Details
Protocol	Bluetooth 4.2 (ESP32)
Max Speed	3 Mbps
Real-World Speed	1–2 Mbps
Range	Up to 10m (Indoors), 50m (Open Space)
Latency	100–200 ms (higher than Wi-Fi)
Best Use Case	Short-range data transfer (passenger count, sensor data)

Methodology

- **GSM Module (SIM800L)**

Feature	Specification
Frequency Bands	Quad-band **850/900/1800/1900 MHz** (Global Compatibility)
GPRS Data Speed	**85.6 kbps** (Class 12)
SIM Card Support	Micro SIM
Power Supply	**3.4V – 4.4V** (Optimal: 4V)
Power Consumption	Idle: **1.5mA**, Transmission: **500mA – 2A**
Communication	UART (Baud Rate: **1200–115200 bps**)
Networking Support	GPRS, TCP/IP, HTTP, MQTT
Antenna	External Antenna Required
SMS & Call Support	Yes
AT Command Support	Standard GSM AT Commands
Temperature Range	-40°C to +85°C
Form Factor	Small (2.5cm × 2.3cm)

SIM800L Pin	ESP32 Pin
VCC (4V)	External 4V Source (Not ESP32 3.3V)
GND	GND
TX (SIM800L)	RX (GPIO16)
RX (SIM800L)	TX (GPIO17)
RST	Not Required (Optional Reset)

AI Model

- **LSTM + SARAMA Hybrid:**
  - SARIMA captures linear patterns and seasonality in time series data.
  - LSTM models complex, non-linear dependencies and long-term trends.
  - Hybrid Integration:
  - Residual Approach: SARIMA forecasts the main trend, and LSTM models the residuals.
  - Direct Approach: Both models make predictions separately, and their outputs are combined (e.g., averaging).

Initial Research and Work Done

- Conducted a comprehensive literature review on AI scheduling, IoT applications in transit, and real-time route optimization.
- Explored on online datasets that are available.
- Explored on creating initial prototypes for sensor integration and edge data processing.
- Designed a basic webpage dashboard.

Progress



Challenges and Risks

Complexity in integrating multiple data sources and hardware components.

Ensuring real-time data reliability and secure transmission.



Scalability of the AI optimization model for larger transit fleets.

Potential issues with hardware ruggedization and power management in field conditions.

Next Steps

- **Finalize sensor integration and edge processing prototypes.**
- **Build the AI model using additional real-time data and further simulations.**
- **Refine and test the cloud-based dashboard.**
- **Prepare for pilot deployment on a selected transit route.**

Conclusion

Key Points:

- Reviewed the background, objectives, and scope of the project.
- Outlined our integrated approach using AI, IoT, and GIS.
- Presented progress and preliminary findings along with identified challenges.

Takeaway:

- A data-driven, real-time scheduling system can significantly improve public transit efficiency, reduce costs, and enhance overall service quality.

Literature Review/ References

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